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Design and platform testing of the compact torus central fueling device for the EAST tokamak

Defeng KONG (孔德峰)¹, Ge ZHUANG (庄革)², Tao LAN (兰涛)^{2,*} , Shoubiao ZHANG (张寿彪)^{3,*}, Yang YE (叶扬)¹, Qilong DONG (董期龙)^{1,4}, Chen CHEN (陈晨)² , Jie WU (吴捷)², Sen ZHANG (张森)², Zhihao ZHAO (赵志豪)^{1,5}, Fanwei MENG (孟凡卫)^{1,5}, Xiaohui ZHANG (张小辉)⁵, Yanqing HUANG (黄艳清)⁶, Fei WEN (文斐)³, Pengfei ZI (訾鹏飞)³ , Lei LI (李磊)³, Guanghai HU (胡广海)³ and Yuntao SONG (宋云涛)^{1,3}

¹ Institute of Energy, Hefei Comprehensive National Science Center, Hefei 230031, People's Republic of China

² University of Science and Technology of China, Hefei 230026, People's Republic of China

³ Institute of Plasma Physics, Chinese Academy of Sciences, Hefei 230031, People's Republic of China

⁴ Shihezi University, Shihezi 832003, People's Republic of China

⁵ School of Computer Science and Information Engineering, Hefei University of Technology, Hefei 230009, People's Republic of China

⁶ Hengyang Normal University, Hengyang 421002, People's Republic of China

E-mail: lantao@ustc.edu.cn and bszhang@ipp.ac.cn

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Abstract

Compact torus (CT) injection is a highly promising technique for the central fueling of future reactor-grade fusion devices since it features extremely high injection velocity and relatively high plasma mass. Recently, a CT injector for the EAST tokamak, EAST-CTI, was developed and platform-tested. In the first round of experiments conducted with low parameter settings, the maximum velocity and mass of the CT plasma were $150 \text{ km}\cdot\text{s}^{-1}$ and $90 \mu\text{g}$, respectively. However, the parameters obtained by EAST-CTI were still very low and were far from the requirements of a device such as EAST that has a strong magnetic field. In future, we plan to solve the spark problem that EAST-CTI currently encounters (that mainly hinders the further development of experiments) through engineering methods, and use greater power to obtain a more stable and suitable CT plasma for EAST.

Keywords: compact torus (CT), central fueling, EAST-CTI, EAST tokamak

(Some figures may appear in colour only in the online journal)

1. Introduction

The China Fusion Engineering Test Reactor (CFETR) has two main goals: to achieve 1 GW steady-state fusion power and realize the tritium self-sufficiency operation [1]. Studies have shown that the direct injection of fuel particles (i.e., deuterium or tritium) into the core plasma region can significantly improve the fusion power and reduce the

requirement for tritium breeding rate (TBR). The low TBR helps in achieving tritium self-sufficiency [2, 3]. However, it is difficult for the commonly-used fueling technologies (such as gas puffing, supersonic molecular beam injection (SMBI) [4]), and cryogenic pellets [5]) to reach the core area due to the premature ablation or ionization caused by the high plasma temperature and density. Compact torus (CT) injection might be performed for the central fueling of a reactor-grade tokamak [6]. CT is a self-contained plasmoid proposed by Alfvén, Lindberg, and Mitlid in 1958 [7–9]. It has a high

* Authors to whom any correspondence should be addressed.

plasma density and velocity; it also has a robust axisymmetric poloidal and toroidal magnetic configuration. The basic principle involves the use of a capacitor bank to discharge and break down the working gas for generating the initial plasma. A strong $J_r \times B_t$ force then accelerates the CT plasma to an extremely high directional velocity (here, J_r denotes the radial current density flowing through the plasma and B_t denotes the toroidal magnetic field excited by the current flowing along the inner electrode). The Ring Accelerator Experiment (RACE) device at Lawrence Livermore National Laboratory (LLNL) has attained the highest CT plasma velocity of approximately $2500 \text{ km}\cdot\text{s}^{-1}$ [6].

In the past few decades, studies on the feasibility of this method in tokamaks and its interaction with the tokamak plasma have received extensive attention. For example, Caltech's ENCORE ($R_{\text{max}} = 0.38 \text{ m}$, $a_{\text{min}} = 0.12 \text{ m}$, $B_t = 0.07 \text{ T}$ [10]), Tdev ($R_{\text{max}} = 0.86 \text{ m}$, $a_{\text{min}} = 0.25 \text{ m}$, $B_t = 1.4 \text{ T}$ [11]), JFT-2M ($R_{\text{max}} = 1.31 \text{ m}$, $a_{\text{min}} = 0.35 \text{ m}$, $B_t = 1 \text{ T}$ [12]), and STOR-M ($R_{\text{max}} = 0.46 \text{ m}$, $a_{\text{min}} = 0.12 \text{ m}$, $B_t = 1 \text{ T}$ [13]) are some of the tokamaks on which researchers applied the method. Besides central fueling, many effects, such as the mitigation of disruption [11], high-confinement (H-) mode transition [13], instability modulation, momentum, and auxiliary heating power injection [14] have also been observed. However, this technique has not been verified in a medium/large tokamak with relatively strong magnetic field. However, the conceptual design of the CT fueller for JT-60U with strong magnetic fields ($\sim 2 \text{ T}$) has been introduced in a previous article [15]. Therefore, we designed and built the first CT-injector system for EAST ($R = 1.85 \text{ m}$, $a = 0.45 \text{ m}$, and $B_{t,\text{max}} = 3 \text{ T}$ [16]), a medium-sized fully superconducting tokamak.

This paper is divided into several parts. In section 2, the design of the EAST-CTI is presented. In section 3, a brief introduction to typical discharge and statistical data is presented. Finally, in section 4, a summary is provided.

2. Device design

2.1. The required CT plasma parameters were initially evaluated

To realize the density feedback control on EAST, particle-fueling by CT was performed to effectively supplement the particle loss of the EAST plasma, and the mass of a single CT plasmoid was estimated based on the following formula:

$$m_{\text{CT}} = \frac{N_T}{\tau_P f_{\text{CT}}} m_D. \quad (1)$$

Here, N_T represents the total particles inside the last closed flux surface of EAST, τ_P represents the particle confinement time, f_{CT} represents the designed CT injection frequency, and m_D represents the atomic mass of deuterium. In the usual high-performance operation of EAST, N_T is approximately 10^{20} ; is approximately 100 milliseconds [17], and f_{CT} is the key as its value directly determines the cost of the device. Based on the capability of the existing pulse switches with

high current load and fast current climb rate, we set f_{CT} as 50 Hz preliminarily [18]. Therefore, according to the above-mentioned formula, the mass of a single CT plasma should reach $220 \mu\text{g}$.

To ensure that the CT plasma penetrates the tokamak's target position, the CT kinetic energy density should exceed the magnetic field energy density in the toroidal direction of the tokamak. The equation can be expressed as:

$$\frac{1}{2} n_{\text{CT}} m_D v_{\text{ct}}^2 > \frac{B_t^2}{2\mu_0}. \quad (2)$$

Here, n_{CT} represents the density of the CT plasma, and B_t represents the toroidal magnetic field strength of tokamak. Considering the port size limitation on the EAST device and the elongated effect of the CT plasma, the estimated CT volume was approximately 10^{-3} m^3 , and the corresponding density (n_{CT}) was approximately $6.7 \times 10^{22} \text{ m}^{-3}$. The toroidal magnetic field strength of EAST in the core region is usually about 2.5 T; thus, the CT velocity (v_{CT}) should be higher than $150 \text{ km}\cdot\text{s}^{-1}$ for the central fueling.

2.2. The dimensions of the device were selected based on the above requirements

Similar to most CT devices, the EAST-CTI system consists of a formation region, a compression region, and an acceleration region. The overall structure is shown in figure 1; the inner electrodes of the formation and the acceleration regions are independent of each other. However, the outer electrode is common.

2.2.1. Formation region. Considering that the outer radius of the formation region ($r_{\text{o_form}}$) directly affects the maximum mass that the CT plasma can obtain, the dimensions of the HIT-CTI ($r_{\text{o_form}}^{\text{HIT-CTI}} = 100 \text{ mm}$, $m_{\text{CT}}^{\text{HIT-CTI}} \sim 0.15 \text{ mg}$ [19, 20]), whose CT mass was the closest to our target, was referred to in our design ($r_{\text{o_form}}^{\text{EAST-CTI}} = 125 \text{ mm}$). The estimation of the inner radius ($r_{\text{i_form}}$) depends on the formation threshold (λ_{th}) of CT plasma calculated using Barnes' theory [21]:

$$\frac{\mu_0 I_{\text{gun}}}{B_{\text{bias}} \pi r_{\text{i_form}}^2} > \lambda_{\text{th}} = \frac{1}{r_{\text{i_form}}} \sqrt{\frac{2}{\ln(r_{\text{o_form}}/r_{\text{i_form}})}}. \quad (3)$$

Here, I_{gun} denotes the current flowing through the formation region and B_{bias} denotes the biased magnetic field. The $r_{\text{i_form}}$ should be below 120.0 mm to form a spheromak configuration plasma with a B_{bias} of 0.1 T and an I_{gun} of 100 kA; it was set to 104.5 mm for the EAST-CTI. The length of the formation region (L_{form}) depends on the gas diffusion. The value of L_{form} must be considerably larger than the value of $(r_{\text{o_form}} - r_{\text{i_form}})$ to prevent the working gas from entering the compression region during diffusion. Here, the designed L_{form} of EAST-CTI was 420.0 mm, i.e., 20 times the value of $(r_{\text{o_form}} - r_{\text{i_form}})$.

2.2.2. Acceleration region. We designed the outer radius of the acceleration region according to the port size on the EAST side ($r_{\text{o_acc}} = 50.0 \text{ mm}$). The evaluation of the inner radius

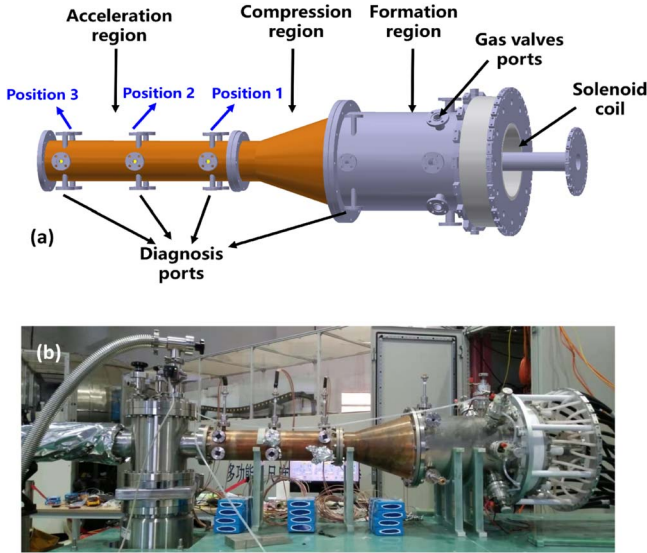


Figure 1. A schematic representation (a) and a photograph (b) of the EAST-CTI.

depends on the simulation of the 1D point model [6]. The result of the point model showed that the energy utilization rate of the power supply improved with the increase in the ratio of the outer radius to the inner radius (r_{o_acc}/r_{i_acc}) [22], i.e., a very low value of r_{i_acc} was favorable. However, a very small value of r_{i_acc} increases the chamber volume, thereby diluting the density and magnetic field strength of the plasma. Additionally, this might also increase the inductance (loop area $\uparrow$) and resistance (cross-sectional area $\downarrow$) of the acceleration region, which is unfavorable for CT acceleration. Moreover, in the small r_{i_acc} case, the toroidal field induced by the discharge current along the acceleration inner electrode may be an uneven distribution, resulting in a large gradient of $\mathbf{J} \times \mathbf{B}$ force in the r direction, and the blow-by effect can occur near the inner electrode, which is unfavorable for CT plasma [23]. Considering the energy utilization rate, density uniformity, and lifetime of the CT plasmoid, we set the r_{o_acc}/r_{i_acc} of EAST-CTI as 2.9 [22]. The calculation of the acceleration region length ($L_{acc} = 500.0$ mm) was also based on the point model, in which the acceleration bank voltage was set to 15 kV and the electrode material was oxygen-free copper with low resistivity.

2.2.3. Compression region. The compression region's taper should be selected carefully. The compression time (τ_{comp}) of the CT plasma must be long enough, i.e., the taper should be as small as possible. A small taper ensures that the plasma undergoes a self-similar compression. It also avoids the exciting shock waves (i.e. the Alfvén waves) that might hinder the compression. However, the taper should not be too small since the compression time must be shorter than its lifetime ($\tau_{Alfvén}^{CT} < \tau_{comp} < \tau_{CT-lifetime}$). According to some studies [24–26], the taper of the EAST-CTI was set to 1:1.8, i.e., the length of the compression region (L_{comp}) was 278.0 mm. Based on this, the estimated $\tau_{Alfvén}^{CT}$ was about a few

microseconds ($\tau_{Alfvén}^{CT} = \frac{L_{comp}}{2 v_{Alfvén}^{CT}} = \frac{L_{comp}}{2 B_{CT} / \sqrt{\mu_0 \rho_{CT}}}$, here, B_{CT} and ρ_{CT} indicate the magnetic field strength and density of the CT plasma, respectively). The estimation from the point model showed that τ_{comp} was approximately a dozen microseconds. Additionally, based on the device size and the Spitzer resistivity of the CT plasma, $\tau_{CT-lifetime}$ was several microseconds [9, 27]. Therefore, the above-mentioned time relationship was satisfied.

The dimensions of these three regions are shown in table 1 for better illustration. Additionally, eight sets of fast electromagnetic valves [28, 29] were spaced evenly around the outer formation electrode (figure 1(a)). These valves were parallel-powered to obtain an angularly uniform gas mixing. A solenoid coil with a radius of 98.0 mm and turns of 135×6 was inside the inner formation electrode to generate a quasi-steady state bias magnetic flux for the formation of CT plasma. Furthermore, 20 diagnostic ports were arranged on the shell of the device to determine the performance of plasma. A photograph of the EAST-CTI is shown in figure 1(b).

2.3. Power supply systems

The power supply systems of the EAST-CTI had four sub-systems: the solenoid, gas valve, formation bank, and acceleration bank. The basic circuit diagram is shown in figure 2. The entire circuit was grounded at a single point to protect the power supply systems from the possible ground loops, and a negative high voltage operation was adopted to reduce the blow-by effect in the experiment [30–34].

- (1) The solenoid power supply provides a relatively stable background magnetic flux; the pulse width of the solenoid current should be long enough for the magnetic penetration and the duration of CT. Here, for EAST-CTI, the selected capacitor of the solenoid system was 500 μ F and the maximum voltage was set to 2 kV. In the actual discharge, when the solenoid voltage was 1.3 kV, the generated current reached 80 A, and the resulting magnetic flux was approximately 5 mWb, which met the bias magnetic flux requirement for CT formation. Additionally, the measured current pulse width was approximately 20 ms, which was considerably longer than the CT duration time (< 0.1 ms) and the magnetic field penetration time (< 0.05 ms) [29]. Therefore, the steady-state magnetic field requirement was satisfied during CT formation and acceleration.
- (2) The gas valve power supply should simultaneously provide a fast-climbing current to the eight sets of valves. Here, the gas valve used was the same one as that on our collaborator KTX-CTI [28, 29]. According to the experiences with KTX-CTI, to achieve a sufficient amount of gas injection, the current flowing through a single gas valve needs to reach 900 A; therefore, a total of 7.2 kA current was required. The selected capacitance of the gas valve system was 250 μ F, and the set maximum voltage was 2 kV. The maximum voltage generated had a peak current of about 8 kA. Additionally,

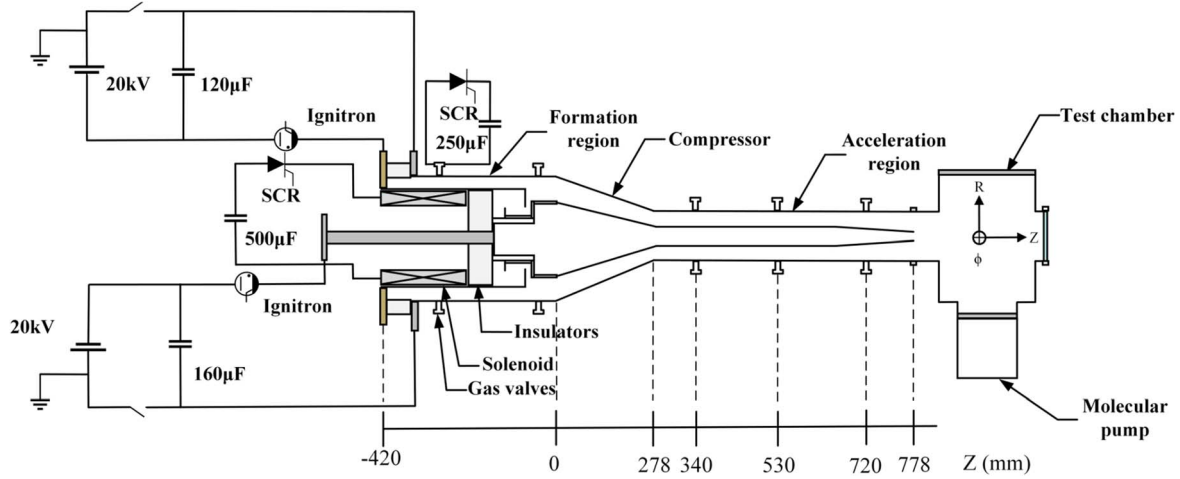


Figure 2. Simplified schematic of EAST-CTI circuit.

Table 1. Dimensions of EAST-CTI.

		Inner radii (mm)	Outer radii (mm)	Length (mm)
Formation region		104.5	125.0	420.0
Compression region	Front	104.5	125.0	278.0
	Behind	17.5	50.0	
Acceleration region		17.5	50.0	500.0

the pulse of the electric current was approximately $150 \mu\text{s}$, and the current climb rate was close to $150 \text{ A} \cdot \mu\text{s}^{-1}$, ensuring the rapid pulse injection of the working gas.

- (3) The power supply of the formation bank consisted of four sets of $30 \mu\text{F}$ coaxial capacitors in parallel. It had stored energy of about 24 kJ. The designed voltage was adjustable from 0 to 20 kV (the corresponding current was from 0 to 400 kA). The loop current exceeded 260 kA with 10 kV discharge and the current climb rate reached $20 \text{ kA} \cdot \mu\text{s}^{-1}$. The acceleration bank consisted of four sets of $40 \mu\text{F}$ coaxial capacitors in parallel, similar to the formation bank; the stored energy was approximately 32 kJ. Additionally, the designed maximum voltage was also 20 kV.

2.4. Magnetic shielding

The mechanism of CT was closely related to the magnetic field. The formation and acceleration of CT plasma might be strongly affected by the strong stray magnetic field [15, 35, 36]. According to the designed injection position of the EAST-CTI, the tail of the acceleration inner electrode was about 2 m away from the center of the EAST plasma. During the EAST experiment, the stray magnetic field strength in this region was several hundred Gauss, as recorded by a gauss meter. At this magnitude, magnetic shielding using

Table 2. Differences in the formation bank length of different devices.

	EAST-CTI	KTX-CTI	HIT-CTI	USCTI	RACE
Length of formation bank (mm)	420	~ 1000 [29]	~ 700 [19]	~ 800 [14]	~ 700 [37]

ferromagnets is sufficient. For ferromagnetic materials, the shielding efficiency can be calculated as $S = \frac{\mu_r d}{D}$, where S indicates the magnetic shielding coefficient, μ_r indicates the relative permeability of the material (10 000 for soft iron), d indicates the thickness of the shield, and D indicates the diameter of the shielded body. For the EAST-CTI with a diameter of $D = 0.10 \text{ m}$ in the acceleration region, the shield thickness d was required to be 0.10 mm for setting the shielding coefficient S at 10.

To summarize, the EAST-CTI is a compact device specially designed for EAST, and its principle is similar to that of most devices. A three-stage structure was used (formation, compression, and acceleration). The circuit connection was the same as that of the HIT-CTI [19, 20], the formation and acceleration banks were grounded at the common outer electrode, respectively (as shown in figure 2), and a negative high voltage operation was adopted. The difference was that we moved the vacuum pump to the end of the acceleration region, as shown in figure 1. The advantage of changes in the position of the pump was that the length of the forming region was shortened (as shown in table 2), thus reducing the loop length, as well as the circuit resistance and inductance. It was also beneficial in reducing the inventory of unionized neutral gas in the forming region.

3. Platform test results

The whole EAST-CTI system was fully constructed in September 2021, after which the platform test experiments were performed. In figure 3, a typical helium plasma discharge is

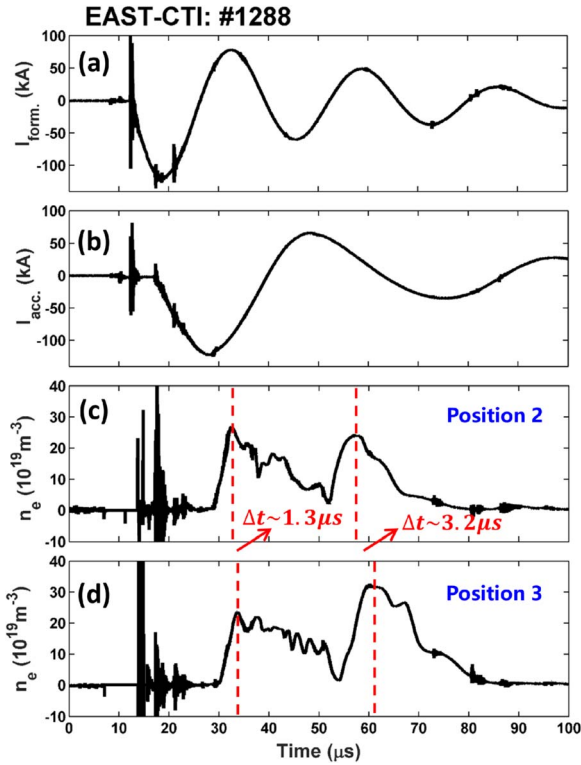


Figure 3. A typical discharge of EAST-CTI (helium). (a), (b) The plasma currents (measured by the Rogowski coils) at the formation and acceleration banks, (c), (d) the electron density (measured by the Langmuir probes) at positions 2 and 3.

shown, where the voltages of the formation and acceleration banks were set to 5.0 and 6.0 kV, respectively. The corresponding peak currents of these two loops were 123 and 125 kA. The turn-on time difference between these two regions was only about 5 μs . This indicated that the plasmoid quickly passed through the formation and compression regions before entering the acceleration region. Two plasmoids were formed in this shot, as shown in figures 3(c) and (d); the plasma velocities estimated using the time-of-flight method were about 150 and 60 $\text{km}\cdot\text{s}^{-1}$. A lower velocity in the second CT plasma might be due to a weak current in the acceleration bank, or probably because the second half cycle of the acceleration current was not suitable for the second CT plasmoid. The mass of the CT plasmoid was estimated by the following formula:

$$m_{\text{CT}} \approx m_{\text{He}} v_{\text{CT}} S \int n_e dt \quad (4)$$

where S and n_e denote the cross-sectional area and line-averaged electron density of the CT plasmoid and m_{He} denotes the atomic mass of helium. The estimated plasma masses of these two plasmoids near the muzzle (position 3, figure 1(a)) were about 30 μg (time = 30–54 μs) and 15 μg (time = 54–80 μs), respectively, as shown in figure 3(d). It is worth pointing out that some inaccuracies exist in the mass estimation. First, the CT plasma exhibited quasi-neutral characteristics, and the temperature was generally lower (~ 10 eV [29]), considerably lower than the second ionization energy of helium atoms (~ 54.5 eV). Therefore, the main ion of CT plasma was considered to be

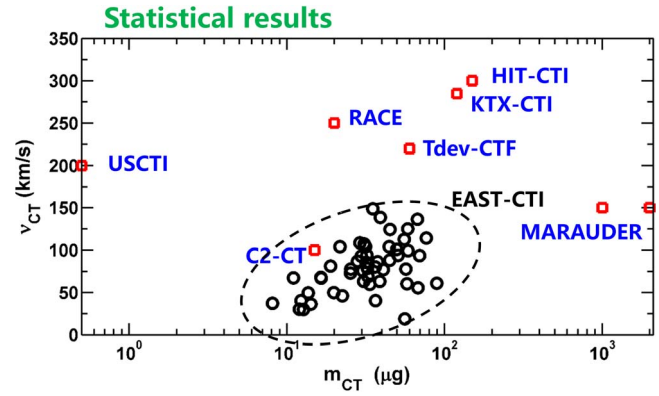


Figure 4. Data on the velocity and mass of the CT plasma obtained from different devices (USCTI [27], RACE [37], C2-CT [38], Tdev-CTF [39], HIT-CTI [12], MARAUDER [25], and KTX-CTI [29]).

Table 3. Basic information and capabilities of #1288.

Voltage U (kV)	5 (U_{form}) and 6 (U_{acc})
Current I (kA)	123 (I_{form}) and 125 (I_{acc})
CT velocity v ($\text{km}\cdot\text{s}^{-1}$)	~ 150 (1st CT) and ~ 60 (2nd CT)
Particle inventory N_{CT} (helium 10^{18})	~ 1.2 (1st CT) and ~ 0.56 (2nd CT)
$N_{\text{CT}}/N_{\text{EAST}}$	$\sim 0.3\%$ (1st CT) and $\sim 0.15\%$ (2nd CT)
CT mass m (μg)	~ 30 (1st CT) and ~ 15 (2nd CT)
CT length L_{CT} (m)	~ 1.5 (1st CT) and ~ 0.7 (2nd CT) at position 3
$L_{\text{CT}}/\text{minor radius of EAST}$	~ 3.2 (1st CT) and ~ 1.5 (2nd CT)
Penetrating magnetic field strength (T)	~ 0.38 (1st CT) and ~ 0.19 (2nd CT)

He^+ , i.e., the ion density was considered to be close to the electron density. Second, the use of average velocity for the estimation might be inaccurate. Finally, the electron density used here was a local value measured by the Langmuir probe, which deviated from the average electron density. However, these approximations probably did not affect the assessment of the CT mass on the magnitude scale. The basic information and the capabilities of this shot are summarized in table 3.

We performed a preliminary scan of the parameter window of EAST-CTI during the first round of platform experiments over three months. The statistical plots are shown in figure 4; the circles represent the EAST-CTI results, and the solid squares represent the typical results from the mainstream CT devices published in other studies. The CT parameters we obtained were relatively low; the maximum velocity and mass of the CT plasma in the EAST-CTI were approximately 150 $\text{km}\cdot\text{s}^{-1}$ and 90 μg , respectively. The low values were due to the relatively lower discharge parameters. The tested highest voltages of the formation and acceleration banks were only 8 kV; the designed maximum voltage reached 20 kV. Additionally, the velocity of the EAST-CTI in this plot was the average value calculated by the Langmuir probes at positions 2 and 3 (the locations are provided in figure 1(a)). The velocity of the muzzle must be higher than the average velocity of positions 2 and 3. Due to the lack of experience in engineering construction, sparks often occur in the host system during

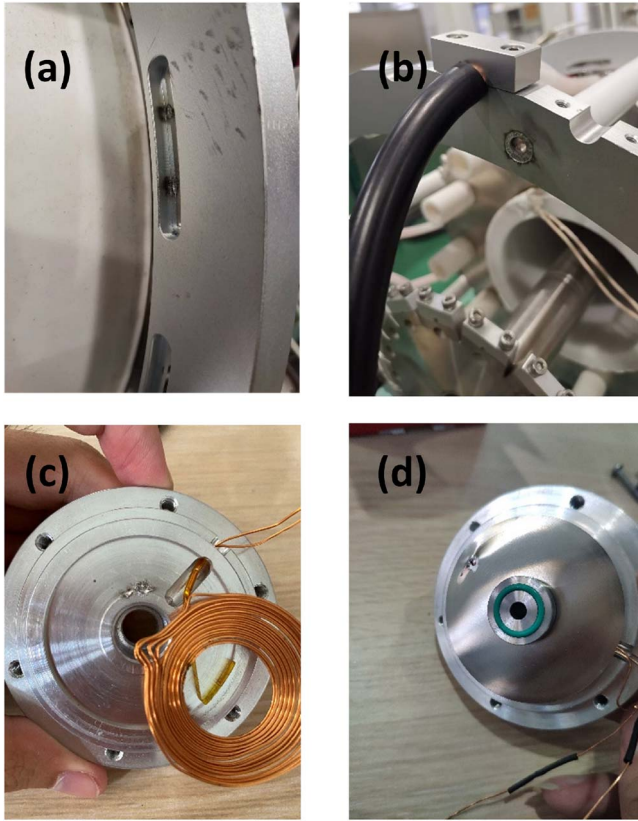


Figure 5. Part of the sparks occurred in EAST-CTI. (a), (b) Connection components of the cable, (c), (d) the internal components of the gas valve.

high-voltage experiments. This might cause the plasma to fail to establish and sometimes damage the power supply systems in a more severe case, which limits the EAST-CTI to being used for conducting experiments with a higher power. The damaged connection components of the circuit and the internal components of the gas valve are shown in figure 5.

4. Discussion

4.1. Modification of the device

To counter the various sparking problems shown in figure 5, a series of device modifications are planned, including:

4.1.1. Connection component modification. As shown in figure 6, the connecting rods (1) were made of aluminum and the connecting discs (2), (3) and (4) were made of stainless steel in the previous structure. Due to the poor crimping, the obvious spark corrosion occurs at these connection positions. In the revised version, it is planned to replace all the connection components to copper, which is beneficial for suppressing sparks and reducing the circuit's resistance. Furthermore, it is planned to increase the number of connecting rods (1) from 8 to 16 to reduce the current that passes through a single rod. The length of the connecting rod has also been reduced from the original 205 mm to 140 mm, and the diameter of the inner disk (4) of the acceleration region has been increased from 50 to 95 mm,

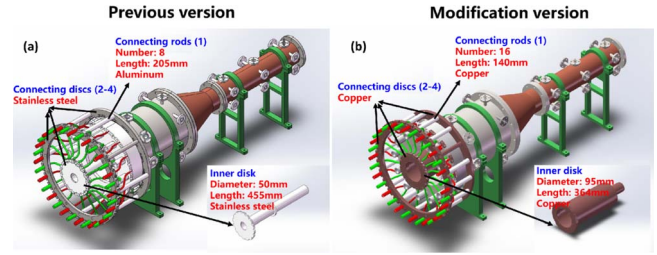


Figure 6. Schematic representation of EAST-CTI in previous version (a) and modification version (b).

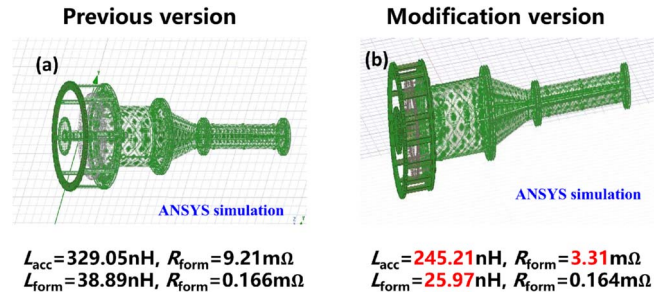


Figure 7. Difference between the circuit parameters simulated by the ANSYS under the two versions.

all of which will help reduce the circuit parameters. Inductance and resistance of the host system are reduced from the previous 329.05 nH/9.21 mΩ to 245.21 nH/3.31 mΩ (acceleration bank) and 38.89 nH/0.166 mΩ to 25.97 nH/0.164 mΩ (formation bank) under this modification, as demonstrated by the ANSYS simulation, as shown in figure 7.

4.1.2. Gas valve modification. The gas valve used here is identical to that of our collaborator KTX-CTI [28, 29], and its internal structure is shown in figure 8. Due to the poor insulation consideration, sparks often occurred within the gas valve, as shown in figures 5(c) and (d) above. After modification, it is planned to install a 0.1 mm thick polyimide insulating sheet among the base (1), the copper coil (2) and the aluminum sheet (3), as shown in figure 8(b).

4.1.3. Solenoid coils modification. Solenoid coils also frequently spark since they are too close to the inner electrode of the formation bank, as shown in figure 9. After modification, it is planned to shorten the diameter of the solenoid coil holder by 5 mm and add a polyimide insulation layer (0.5 mm thick) between the holder and the solenoid coils.

4.2. CT density

According to the experience of multiple devices around the world, it is very difficult to obtain an electron density as high as the designed value of $6.7 \times 10^{22} \text{ m}^{-3}$. However, it is worth pointing out that as an experimental device, EAST does not rely on CT to achieve its particle balance, which would greatly reduce the demand for CT density. The density requirement can therefore be reduced by increasing the injection velocity of CT appropriately. According to the formula for magnetic field penetration of CT, if the velocity of CT plasma could be

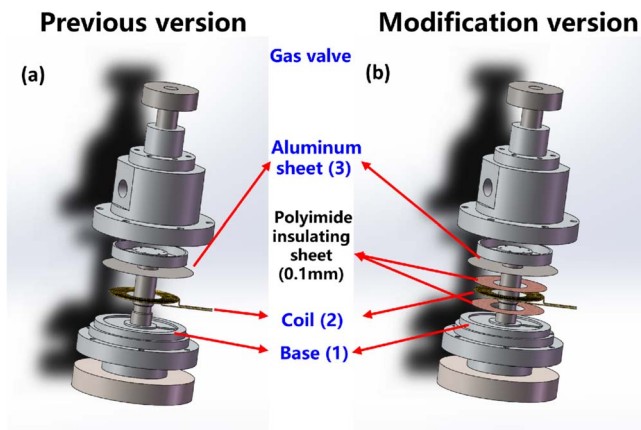


Figure 8. Schematic representation of the gas valve in previous version (a) and modification version (b).

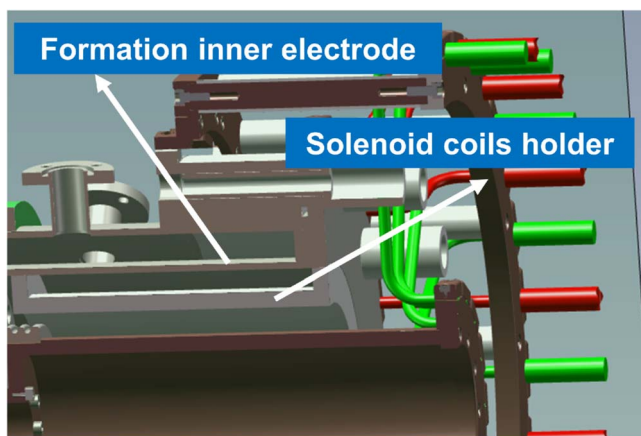


Figure 9. Schematic diagram of the relative position of the inner electrode of the formation bank and the solenoid coils holder.

increased to $300 \text{ km}\cdot\text{s}^{-1}$, the demand density would drop to the order of $\sim 1 \times 10^{22} \text{ m}^{-3}$, which seems to be a relatively easy target to achieve. In addition, by increasing the injection velocity it also offers the possibility of long distance traverses. As mentioned above, the position reserved for the CT is far away from the main plasma of the tokamak due to the space constraints at the periphery of EAST, with the tail of the acceleration inner electrode being approximately 2 m long from the center of EAST. It will take the CT in the order of $13 \mu\text{s}$ to travel it with velocity at $150 \text{ km}\cdot\text{s}^{-1}$ and $7 \mu\text{s}$ at $300 \text{ km}\cdot\text{s}^{-1}$. Considering that the lifetime of CT is usually on the timescale of a few tens of μs [9, 23], the increased injection velocity will save the travel and thus provide ample time for the acceleration and the subsequent reconnection of the magnetic field. The next step in the experimental exploration of EAST-CTI will focus on optimizing the device structure and the discharge parameters to achieve higher injection velocity.

5. Summary

To summarize, a newly-developed CT injector for the medium-sized fully superconducting tokamak EAST was developed and tested. The dimensions of the device, power

supplies, the first round of low parameter platform experiments and the plans for the modifications were described in this paper. It is worth pointing out that the parameters obtained by EAST-CTI were very low and were far from the values required to achieve any core fueling in the EAST strong magnetic field environment. In future, the EAST-CTI Research & Development team aims to perform higher parameter scanning experiments and actively solve the engineering problems currently encountered.

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ORCID iDs

Tao LAN (兰涛) <https://orcid.org/0000-0002-2785-5178>
 Chen CHEN (陈晨) <https://orcid.org/0000-0002-1072-4999>
 Pengfei ZI (訾鹏飞) <https://orcid.org/0000-0001-5393-135X>

References

- [1] Song Y T *et al* 2014 *IEEE Trans. Plasma Sci.* **42** 503
- [2] Xie H *et al* 2020 *Nucl. Fusion* **60** 046022
- [3] Jackson G L, Chan V S and Stambaugh R D 2013 *Fusion Sci. Technol.* **64** 8
- [4] Wang E Y 2001 *Plasma Sci. Technol.* **3** 673
- [5] Rêgo I, da S *et al* 2006 *Fusion Eng. Des.* **81** 2649
- [6] Hammer J H *et al* 1988 *Phys. Rev. Lett.* **61** 2843
- [7] Taylor J B 1986 *Rev. Mod. Phys.* **58** 741
- [8] Parks P B 1988 *Phys. Rev. Lett.* **61** 1364
- [9] Perkins L J, Ho S K and Hammer J H 1988 *Nucl. Fusion* **28** 1365
- [10] Brown M R and Bellan P M 1992 *Nucl. Fusion* **32** 1125
- [11] Raman R *et al* 1994 *Phys. Rev. Lett.* **73** 3101
- [12] Ogawa T *et al* 1999 *Nucl. Fusion* **39** 1911
- [13] Sen S *et al* 2002 *Phys. Rev. Lett.* **88** 185001
- [14] Liu D Z *et al* 2006 *Nucl. Fusion* **46** 104
- [15] Raman R and Itami K 2000 *J. Plasma Fusion Res.* **76** 1079
- [16] Ninomiya H *et al* 2002 *Fusion Sci. Technol.* **42** 7
- [17] Yu Y W *et al* 2019 *Nucl. Fusion* **59** 126036
- [18] Loree D L *et al* 1991 *IEEE Trans. Electron Devices* **38** 720
- [19] Fukumoto N *et al* 2004 *Fusion Eng. Des.* **70** 289
- [20] Fukumoto N *et al* 2006 *Fusion Eng. Des.* **81** 2849
- [21] Barnes C W *et al* 1990 *Phys. Fluids B: Plasma Phys.* **2** 1871
- [22] Dong Q L *et al* 2022 *Plasma Sci. Technol.* **24** 025103
- [23] Miyazawa J *et al* 2001 *Fusion Eng. Des.* **54** 1
- [24] Hartman C W and Hammer J H 1982 *Phys. Rev. Lett.* **48** 929

- [25] Degnan J H *et al* 1993 *Phys. Fluids B: Plasma Phys.* **5** 2938
- [26] Hameiri E, Stevens D C and Grossmann W 1985 *Phys. Fluids* **28** 590
- [27] Xiao C, Hirose A and Sen S 2004 *Phys. Plasmas* **11** 4041
- [28] Thomas J C *et al* 1993 *Rev. Sci. Instrum.* **64** 1410
- [29] Chen C *et al* 2022 *Plasma Sci. Technol.* **24** 045102
- [30] Fishman F J and Petschek H 1962 *Phys. Fluids* **5** 632
- [31] Keck J C 1962 *Phys. Fluids* **5** 630
- [32] Rabiński M and Zdunek K 2007 *Surf. Coat. Technol.* **201** 5438
- [33] Klar T A *et al* 2014 *Phys. Scr.* **2014** 014049
- [34] Yu X *et al* 2020 *Acta Phys. Sin.* **69** 035202 (in Chinese)
- [35] Yee J and Bellan P M 1998 *Nucl. Fusion* **38** 711
- [36] Raman R and Gierszewski P 1998 *Fusion Eng. Des.* **39–40** 977
- [37] Hammer J H *et al* 1991 *Phys. Fluids B: Plasma Phys.* **3** 2236
- [38] Asai T *et al* 2017 *Nucl. Fusion* **57** 076018
- [39] Raman R *et al* 1997 *Nucl. Fusion* **37** 967